

Math 142, S14
Exam 1

Section:

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	18	
2	<i>18</i>	
3	20	
4	20	
5	<i>24</i>	
Total	100	

Good Luck !

Problem 1. (18 points) Evaluate the following integrals.

(a) $\int e^x \sqrt{1+e^x} dx$

let $u = 1+e^x$, then $du = e^x dx$

$$\int e^x \sqrt{1+e^x} dx = \int \sqrt{u} du$$

$$= \frac{2}{3} u^{3/2} + C$$

$$= \frac{2}{3} (1+e^x)^{3/2} + C$$

(b) $\int \frac{x}{1+x^4} dx$

let $u = x^2$, then $du = 2x dx$

$$\int \frac{x}{1+x^4} dx = \int \frac{1}{1+u^2} \cdot \frac{1}{2} du$$

$$= \frac{1}{2} \arctan u + C$$

$$= \frac{1}{2} \arctan(x^2) + C$$

(c) $\int_0^{\pi/4} \tan^2 \theta \sec^4 \theta d\theta$

let $u = \tan \theta$, then $du = \sec^2 \theta d\theta$.

$$\theta = 0 \rightarrow u = \tan 0 = 0$$

$$\theta = \pi/4 \rightarrow u = \tan(\pi/4) = 1.$$

$$\int_0^{\pi/4} \tan^2 \theta \sec^4 \theta d\theta = \int_0^{\pi/4} \tan^2 \theta (1+\tan^2 \theta) \sec^2 \theta d\theta$$

$$= \int_0^1 u^2 (1+u^2) du = \int_0^1 u^2 + u^4 du$$

$$= \left[\frac{u^3}{3} + \frac{u^5}{5} \right]_{u=0}^{u=1}$$

$$= \left[\frac{1}{3} + \frac{1}{5} \right] - [0] = \frac{8}{15}$$

Problem 2. (18 points) Evaluate the following integrals.

(a) $\int x e^{-3x} dx$

$$u = x, dv = e^{-3x} dx$$

$\downarrow$

$$du = dx, v = -\frac{1}{3} e^{-3x}$$

$$\Rightarrow \int x e^{-3x} dx = x \cdot \left(-\frac{1}{3} e^{-3x}\right) - \int -\frac{1}{3} e^{-3x} dx$$

$$= -\frac{x}{3} e^{-3x} - \frac{1}{9} e^{-3x} + C$$

$$= e^{-3x} \left(-\frac{x}{3} - \frac{1}{9}\right) + C$$

(b) $\int \sin \sqrt{t} dt$

let $x = \sqrt{t}$, then $x^2 = t$, $2x dx = dt$

$$\int \sin \sqrt{t} dt = \int 2x \sin x dx$$

$$= 2 \left[-x \cos x - \int (-\cos x) dx \right]$$

$$= -2x \cos x + 2 \int \cos x dx$$

$$= -2x \cos x + 2 \sin x + C$$

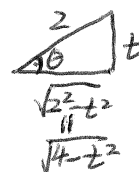
$$= -2\sqrt{t} \cos \sqrt{t} + 2 \sin \sqrt{t} + C$$

Problem 3. (20 points) Evaluate the following integrals.

(a) $\int \sqrt{4-t^2} dt$

let $t = 2\sin\theta$, then $dt = 2\cos\theta d\theta$.
 $\downarrow$
 $\theta = \arcsin(t/2)$

$$\begin{aligned} \int \sqrt{4-t^2} dt &= \int 2\cos\theta \cdot 2\cos\theta d\theta = \int 4\cos^2\theta d\theta \\ &= \int 4 \cdot \frac{1}{2} (1 + \cos 2\theta) d\theta = \int 2(1 + \cos 2\theta) d\theta \\ &= 2(\theta + \frac{1}{2}\sin 2\theta) + C \\ &= 2\theta + \sin 2\theta + C \\ &= 2\arcsin(t/2) + 2\sin\theta\cos\theta + C \\ &= 2\arcsin(t/2) + 2 \cdot \frac{t}{2} \cdot \frac{\sqrt{4-t^2}}{2} + C \\ &= 2\arcsin(t/2) + \frac{t\sqrt{4-t^2}}{2} + C \end{aligned}$$



(b) $\int \frac{x+3}{x(x^2-1)} dx$

$$\begin{aligned} x(x^2-1) &= x(x-1)(x+1) \\ \Rightarrow \frac{x+3}{x(x^2-1)} &= \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} \\ &= \frac{A(x^2-1)}{x(x^2-1)} + \frac{Bx(x+1)}{x(x^2-1)} + \frac{Cx(x-1)}{x(x^2-1)} \\ &= \frac{(Ax^2-A) + Bx^2+Bx + Cx^2-Cx}{x(x^2-1)} = \frac{(A+B+C)x^2 + (B-C)x - A}{x(x^2-1)} \\ \Rightarrow \begin{cases} A+B+C=0 \\ B-C=1 \\ -A=3 \end{cases} &\Rightarrow \begin{cases} A=-3 \\ B=2 \\ C=1 \end{cases} \\ \Rightarrow \int \frac{x+3}{x(x^2-1)} dx &= \int -\frac{3}{x} + \frac{2}{x-1} + \frac{1}{x+1} dx \\ &= -3\ln|x| + 2\ln|x-1| + \ln|x+1| + C \end{aligned}$$

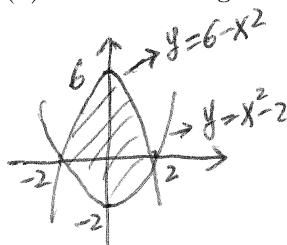
Problem 4. (20 points) Evaluate each of the following improper integrals or show that it is divergent.

$$\begin{aligned}
 \text{(a)} \quad & \int_2^{+\infty} \frac{dy}{y(\ln y)^2} \\
 &= \lim_{t \rightarrow +\infty} \int_2^t \frac{1}{y(\ln y)^2} dy \\
 &= \lim_{t \rightarrow +\infty} \left[-\frac{1}{\ln y} \right]_{y=2}^t \\
 &= \lim_{t \rightarrow +\infty} \left[-\frac{1}{\ln t} + \frac{1}{\ln 2} \right] = 0 + \frac{1}{\ln 2} = \frac{1}{\ln 2} \\
 &\Rightarrow \text{convergent}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad & \int_2^{10} \frac{dx}{\sqrt[3]{10-x}} \\
 &= \lim_{t \rightarrow 10^-} \int_2^t \frac{1}{\sqrt[3]{10-x}} dx \\
 &= \lim_{t \rightarrow 10^-} \left[-\frac{3}{2} (10-x)^{\frac{2}{3}} \right]_{x=2}^{x=t} \\
 &= \lim_{t \rightarrow 10^-} \left[-\frac{3}{2} (10-t)^{\frac{2}{3}} \right] + 6 \\
 &= 6 \\
 &\Rightarrow \text{convergent.}
 \end{aligned}$$

Problem 5. (24 points)

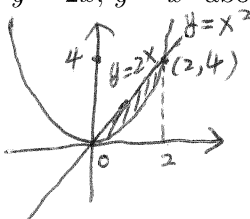
(a) Sketch the region enclosed by the curves $y = 6 - x^2$, $y = x^2 - 2$ and find its area.



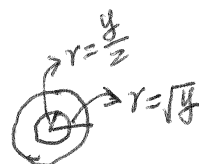
$$\begin{cases} y = 6 - x^2 \\ y = x^2 - 2 \end{cases} \Rightarrow \begin{aligned} 6 - x^2 &= x^2 - 2 \\ 2x^2 &= 8 \\ x^2 &= 4 \\ x &= \pm 2 \end{aligned}$$

$$\begin{aligned} A &= \int_{-2}^2 (6 - x^2) - (x^2 - 2) dx \\ &= \int_{-2}^2 8 - 2x^2 dx \\ &= \left[8x - \frac{2}{3}x^3 \right]_{x=-2}^{x=2} \\ &= \left(16 - \frac{16}{3} \right) - \left(-16 + \frac{16}{3} \right) \\ &= \frac{32}{3} + \frac{32}{3} = \frac{64}{3} \end{aligned}$$

(b) Find the volume of the solid obtained by rotating the region bounded by the curves $y = 2x$, $y = x^2$ about the y -axis.



$$\begin{cases} y = 2x \\ y = x^2 \end{cases} \Rightarrow \begin{aligned} 2x &= x^2 \\ x^2 - 2x &= 0 \\ x(x - 2) &= 0 \\ x &= 0 \text{ (} y=0 \text{), or } x=2 \text{ (} y=4 \text{)} \end{aligned}$$



Method 1: $V = \int_0^2 2\pi x (2x - x^2) dx$

$$\begin{aligned} &= \int_0^2 2\pi (2x^2 - x^3) dx \\ &= 2\pi \left[\frac{2}{3}x^3 - \frac{x^4}{4} \right]_{x=0}^{x=2} \\ &= 2\pi \left[\left(\frac{16}{3} - \frac{16}{4} \right) - 0 \right] \\ &= 2\pi \cdot \frac{4}{3} \\ &= \frac{8}{3}\pi \end{aligned}$$

Method 2:

$$\begin{aligned} V &= \int_0^4 \pi (\sqrt{y})^2 - \pi \left(\frac{y}{2} \right)^2 dy \\ &= \int_0^4 \pi \left(y - \frac{y^2}{4} \right) dy \\ &= \pi \left[\frac{y^2}{2} - \frac{y^3}{12} \right]_{y=0}^{y=4} \\ &= \pi \left[\left(\frac{16}{2} - \frac{64}{12} \right) - 0 \right] \\ &= \frac{8}{3}\pi \end{aligned}$$